Sun Ju Lee

ISyE Main 405, 755 Ferst Dr. Atlanta, GA 30332
julee@gatech.edu sjulee.github.io

EDUCATION

Georgia Institute of Technology, Atlanta, GA 8/2019 – Present
H. Milton Stewart School of Industrial and Systems Engineering
Ph.D., Operations Research (Expected 2025)
Advisor: Gian-Gabriel Garcia
Thesis Committee: Lauren Steimle, Weijun Xie, Nicoleta Serban, Sheree Boulet
M.S., Operations Research 5/2022

Dartmouth College, Hanover, NH 9/2014 – 6/2018
B.A., Engineering Sciences
B.E., Mechanical Engineering

PUBLICATIONS

Published and Accepted Papers:

[P2] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet (2023). Interpretable Machine Learning to Predict Adverse Perinatal Outcomes: Examining Marginal Predictive Value of Risk Factors During Pregnancy. *American Journal of Obstetrics & Gynecology MFM*, 101096.

[P1] **S. J. Lee**, X. Gong, and G.-G. P. Garcia (2024). Modified Monotone Policy Iteration for Interpretable Policies in Markov Decision Processes and the Impact of State Ordering Rules. *Annals of Operations Research* (Accepted on July 9, 2024; Preprint).

Working Papers:

[W3] **S. J. Lee** and G.-G. P. Garcia (2024). A Data-driven Optimization Approach to Designing Parsimonious, Interpretable, and Equitable Treatment Guidelines.

[W2] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet (2024). Composite Clustered Multi-Task Learning for Prediction of Adverse Pregnancy Outcomes.

[W1] **S. J. Lee** and G.-G. P. Garcia (2024). A Tolerance-based Approach to Lexicographic Multi-Objective Markov Decision Processes.

BOOK CHAPTERS

[B1] **S. J. Lee**, H. S. Pandey, and G.-G. P. Garcia (2023). Designing Interpretable Machine Learning Models Using Mixed Integer Programming. In: Pardalos, P.M., Prokopyev, O.A. (eds) *Encyclopedia of Optimization.* Springer, Cham.

PRESENTATIONS

Conference Presentations:

[C2] Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes.
– INFORMS Annual Meeting, Phoenix, AZ 10/23

– INFORMS Healthcare Conference, Toronto, ON, Canada 7/23

[C1] A Tolerance-based Approach to Lexicographic Multi-Objective Markov Decision Processes.
- INFORMS Annual Meeting, Indianapolis, IN 10/22
- SMDM Annual Meeting, Seattle, WA 10/22

Invited Seminar Presentations:

1. Interpretable Machine Learning for Adverse Pregnancy Outcomes. 9/23
   School of Industrial and Systems Engineering, University of Oklahoma

HONORS AND AWARDS

Finalist, SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments* 2022
President's Fellow, Georgia Institute of Technology 2019 – 2023
Virtual Annual Meeting & Membership Scholarship, SMDM 2021
Thayer Scholar, Dartmouth College 2014 – 2018

STUDENTS MENTORED

Xingyu Gong BS IE '24 7/2022 – 4/2024
Nathan Grodzinsky BS IE '24 8/2023 – 5/2024

TEACHING EXPERIENCE

Teaching Assistant
Georgia Institute of Technology, Atlanta, GA 8/2019 – 12/2021
- ISyE 3133 Engineering Optimization (undergraduate)
- ISyE 6669 Deterministic Optimization (graduate)
- ISyE 6661 Linear Optimization (graduate)
- ISyE 4134/8813 Constraint Programming (undergraduate/graduate)

Dartmouth College, Hanover, NH 6/2016 – 11/2017
- ENGS 76 Machine Engineering
- ENGS 2 Integrated Design: Engineering, Architecture, and Building Technology
- ENGS 25 Introduction to Thermodynamics

Graduate Tutor
Georgia Institute of Technology, Atlanta, GA 1/2020 – 4/2020
- ISyE 2027 Probability with Applications

LEADERSHIP AND SERVICE ACTIVITIES

Volunteer 6/2024
iExperience Summer Camp, Georgia Tech

Diversity, Equity, and Inclusion Committee Student Member 7/2023 – Present
H. Milton Stewart School of Industrial and Systems Engineering, Georgia Tech

Graduate Research Mentor 5/2022 – 7/2022
Summer Undergraduate Research in Engineering/Sciences Program, Georgia Tech

Teaching Assistant 5/2020 – 6/2020
Seth Bonder Camp in Computational and Data Science for Engineering, Georgia Tech